Samaranjan Manjhi

C/C++ Systems & Cyber Security Software Developer

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SUMMARY

C/C++ Systems and Cyber Security Engineer with **3+ years** of experience building production-grade endpoint security tools, kernel monitors, binary integrity frameworks, and Data Loss Prevention (DLP) systems for **macOS and Linux**. Deep expertise in Endpoint Security Framework (ESF), Elliptic Curve cryptographic signing, execution gating, system call hooking, Qt GUI development, and cross-platform C++ architecture. Delivered **10+ enterprise security modules** deployed globally across macOS and Linux endpoints for enterprise antivirus suites.

TECHNICAL SKILLS

Languages: C, C++, Bash Scripting, Relational SQL
Frameworks: Qt Framework, Endpoint Security Framework (ESF), Extended Berkeley Packet Filter, System Audit, Core Foundation
Security: Cyber Security Architecture, Elliptic Curve Cryptography, Sodium Cryptography Library, OpenSSL Library, Packet Filter Firewall
Tools: Git, CMake, Make, Xcode, GDB, Valgrind, KDB/KGDB, Ftrace/DTrace, Kdump, SQLite, spdlog, XML Parsers, Boost, Nano
Core CS: Data Structures, Multithreading, Inter-Process Communication, Operating System Internals, Linux, macOS

PROFESSIONAL EXPERIENCE

Systems & Security Software Engineer | MicroWorld Software Services Pvt. Ltd. — Mumbai *Aug 2023 – Present*
- Designed and delivered **10+ production security modules** spanning kernel monitors, DLP engines, binary integrity systems, firewall rule generators, and cloud backup agents — deployed globally across enterprise Linux and macOS systems running security products.
- Constructed high-performance cross-platform security daemons in C/C++, integrating Linux kernel modules, macOS ESF event loops, and POSIX Unix domain socket IPC to reduce system memory overhead by 25% and improve event detection throughput by 40%.

KEY PROJECTS

MW Guardian — Binary Integrity & Execution Gating | C/C++, Linux Kernel, Cryptographic Verification *Feb 2026 – Present*
- Implemented execution gating via **kernel permission events** to intercept binary execution at kernel level, synchronously validating cryptographic signatures and terminating untrusted binaries prior to execution.
- Architected a multi-key cryptographic signing pipeline supporting key rotation while retaining legacy public keys to preserve signature validity for previously signed binaries.
- Created startup self-verification routines where guardian and verifier components validate their own binary signatures prior to initiating the main event loop, eliminating bootstrap trust gaps.

Linux Ransomware Defense — Kernel Module | C, Linux Kernel, Syscall Hooking, Behavior Tracking *Dec 2025 – Feb 2026*
- Intercepted file rename, execution, and open system calls using kernel tracing to detect ransomware behavior via threshold-based bulk rename tracking and suffix pattern matching, terminating offending processes and logging security events.
- Enhanced driver stealth by hiding kernel module entries from the active driver list post-load and restricting module unload operations through a password-authenticated system interface.
- Provided runtime configurability at initialization time, allowing dynamic updates to max-rename thresholds, behavior detection toggles, and passphrases without requiring kernel re-compilation.

Printer Control & Data Loss Prevention (DLP) | macOS, C/C++, ESF, Print Service Libraries, Xcode *Oct 2025 – Jan 2026*
- Designed a macOS DLP driver utilizing ESF and print spool monitoring to inspect print files for sensitive information, cancelling unauthorized print jobs prior to printing.
- Implemented application-level process whitelisting and blacklisting enforced via ESF process identity, bypassing clean content for trusted applications and blocking unauthorized applications outright.
- Archived scanned print jobs alongside result metadata to maintain audit trails for compliance reporting and post-event forensic investigations.

File Integrity Monitoring (FIM) | macOS, C/C++, File System Events, SQLite Database, OpenSSL *Jun 2025 – Aug 2025*
- Built a file system event daemon storing cryptographic file hashes, permissions, file sizes, and timestamps in SQLite, emitting real-time security alerts upon unauthorized path modifications, renames, or deletions.
- Established baseline state snapshots on initial startup and executed diff comparisons on subsequent runs to output structured security event logs across configurable watch paths.

Content Scanner — DLP Engine & CLI | C/C++, Regex Engine, Optical Character Recognition, Archive Libraries *Sep 2024 – May 2025*
- Constructed a multi-threaded DLP scanner supporting **15+ file formats** via regular expression parsing and optical character recognition, packaged as a shared library and CLI utility.
- Developed a high-performance Unix domain socket server variant allowing clients to stream file paths and retrieve scan results without re-initializing regex engines, reducing per-scan latency by 60%.

macOS Firewall — Packet Filter Rule Engine | C/C++, Firewall Control, XML Parsers, Network Routing *Sep 2025*
- Generated and applied dynamic firewall rules from XML policies, supporting zone-based access control, trusted MAC address enforcement, Trojan IP blocklists, IP range arithmetic, and live policy reloads without system reboots.
- Parsed multi-zone XML rule sets into anchor configurations, automatically discovering active network interfaces and binding rules dynamically at runtime.

COMPETITIVE PROGRAMMING & PROBLEM SOLVING

Solved Data Structures and Algorithms problems (Arrays, Dynamic Programming, Sliding Window, Graphs, Memory Management) across online judges: **LeetCode | GeeksForGeeks | Coding Ninjas**

EDUCATION

Bachelor of Engineering — Computer Engineering
Shree L.R. Tiwari College of Engineering, Mumbai University — 2019 – 2023 — CGPA: **8.56 / 10**